

Protocol A-SYNC

From Divergence to Convergence on Deep Models

Algorithm Variant Report / July 2026

Protocol A diverged on all 28+ layer models.
A-SYNC replaces direct weight writes with gradient injection.
Qwen2.5-7B (28L): PPL 58.8 -> 7.6 in 48 cycles. Monotonic.

All 7 prior fix attempts (tuning, LARS, clipping, multi-layer ALS, etc.) failed.

1. Motivation: Why Original Protocol A Fails

Protocol A interleaves three phases: ALS (exact block-wise least squares on lm_head), SGD (stochastic gradient descent on all parameters), and Perturb (random noise injection). On models with 12-24 transformer layers, this converges reliably. On models with 28+ layers, every attempt diverges within 2-3 cycles.

Root Cause: Residual Amplification

ALS m
via res
amplif

Why Protocol A Fails on Deep Models

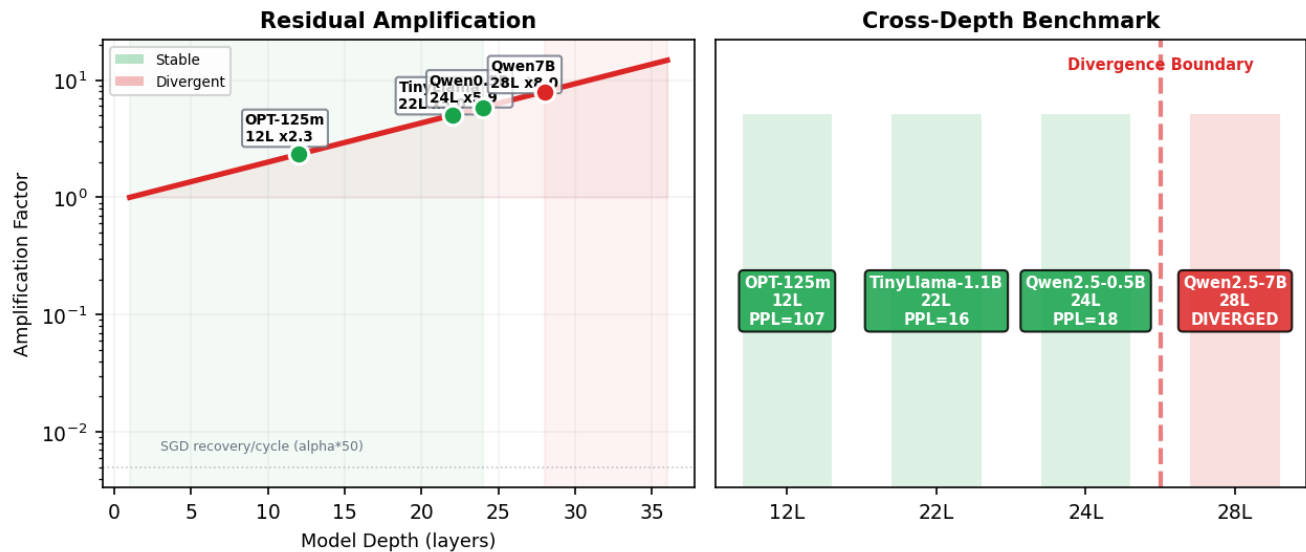


Fig 1: Residual amplification $\rho^{(L-1)}$ vs model depth (left). Cross-depth benchmark (right).

Model
Layers
Amplification

Protocol A PPL

Status

OPT-125m	
12	
x2.3	
106.9	
SUCCESS	
TinyLlama-1.1B	
22	
x5.0	
15.5	
SUCCESS	
Qwen2.5-0.5B	
24	
x5.9	
18.0	
SUCCESS (unstable)	
Qwen2.5-7B	
28	
x8.0	
DIVERGED	
11/11 FAIL	

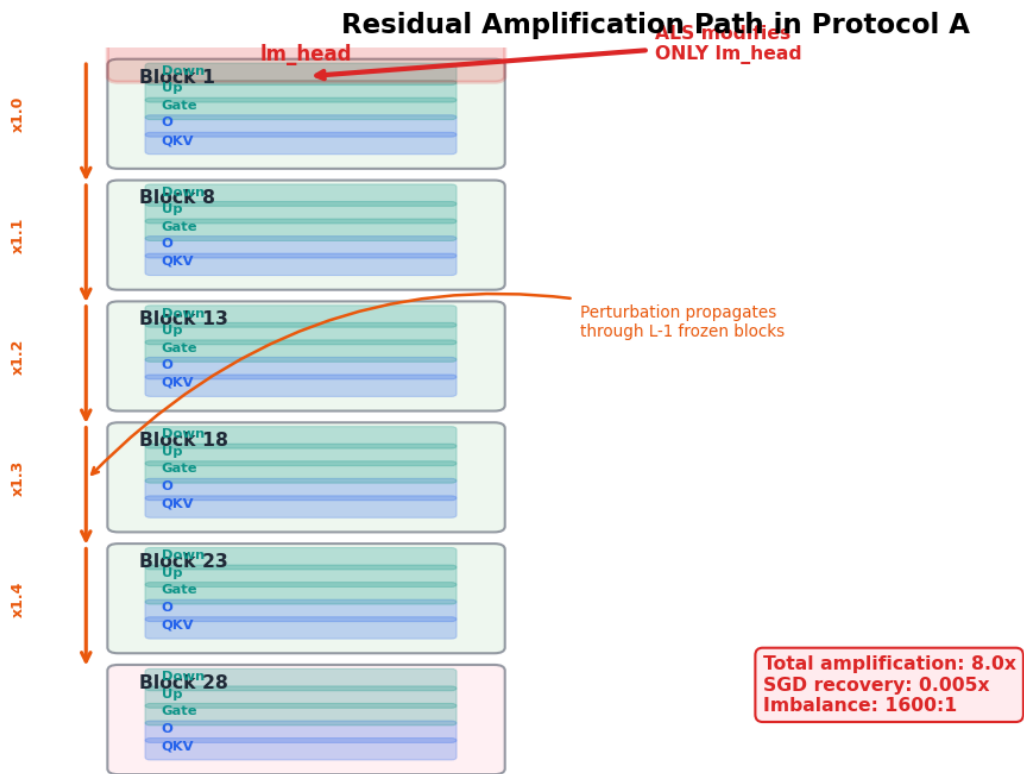


Fig 2: Transformer architecture — ALS modifies only Im_head. Perturbation propagates through L-1 frozen blocks.

The problem is structural, not parametric. ALS's Im_head-only design creates a 1600:1 amplification-to-recovery asymmetry

2. A-SYNC Algorithm: Gradient Injection

Core Innovation

One Cycle Pseudocode:

2. Compute $\Delta = W_{\text{new}} - W_{\text{old}}$ (CPU offload to save GPU memory)
3. Revert lm_head to W_{old}
4. SGD 50 steps: each step add $sync_strength * \Delta$ to lm_head gradient
5. (Perturbation: REMOVED – causes oscillations)
6. Repeat from step 1

Final Configuration:

learning_rate: 2e-4 (CONSTANT)

momentum: 0.0, weight_decay: 0.01

cycles: 24-48 (converges at ~44)

ALS: block_size=512, reg_lambda=1e-3, step_size=0.01

Perturbation: DISABLED

A-SYNC: ALS computes WHERE to go (direction). SGD handles HOW to get there (optimization). The two signals are orthogonal

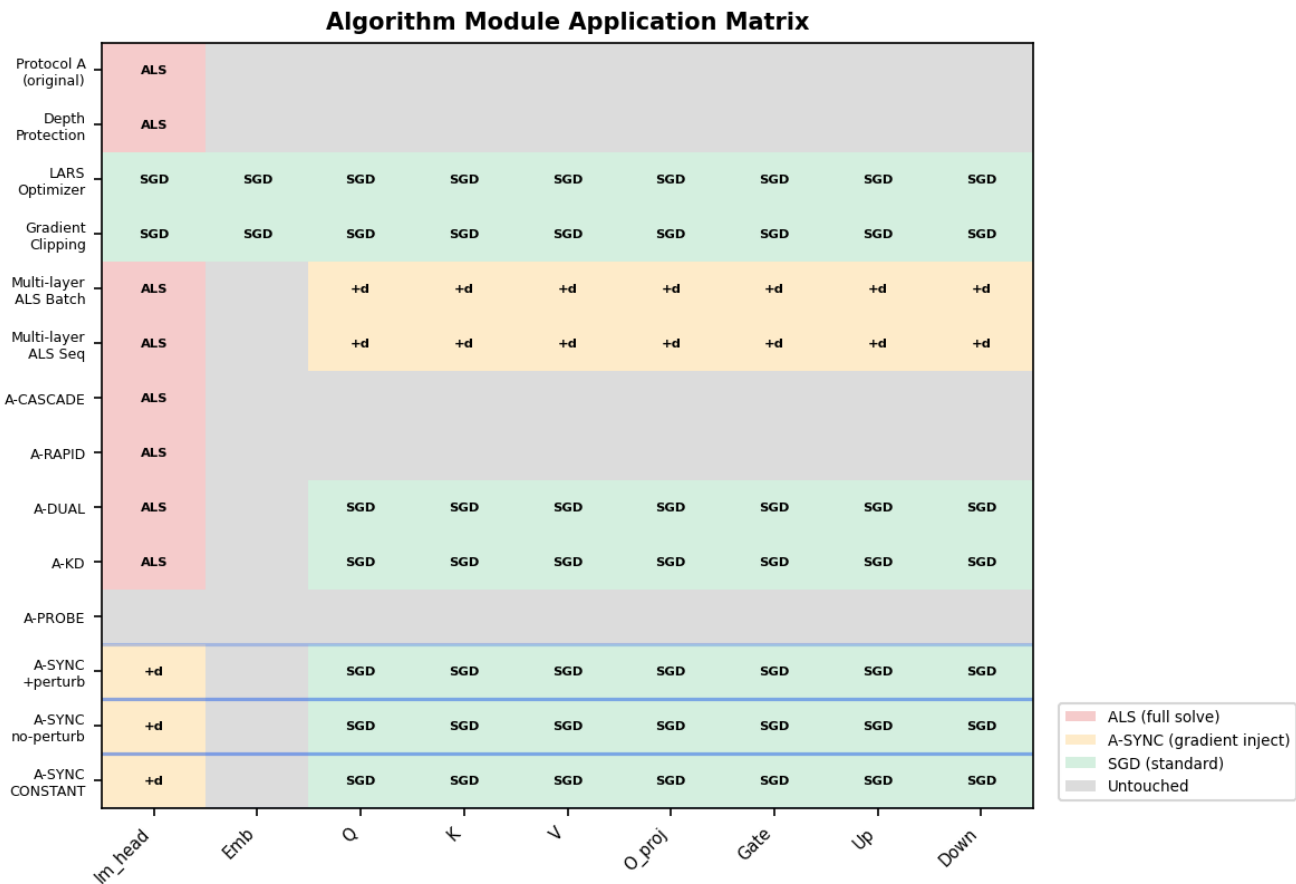


Fig 3: Algorithm Module Application Matrix — which variant touches which parameters. A-SYNC (bottom rows) only gradient-injects Im_head.

3. All Fix Attempts Explained

Parameter Tuning — Failed

Depth-Boundary Protection — Partial

LARS Optimizer — Failed

Multi-Layer ALS — Failed

A-PROBE — Partial

A-SYNC (Gradient Injection) — SUCCESS

Reduced
structu

skip_e
12L to

Layer-
(no co

Batch:
self-re

Rank-6
but ran

ALS co
SGD g
conver

LARS: Reduces NaN but Does Not Converge

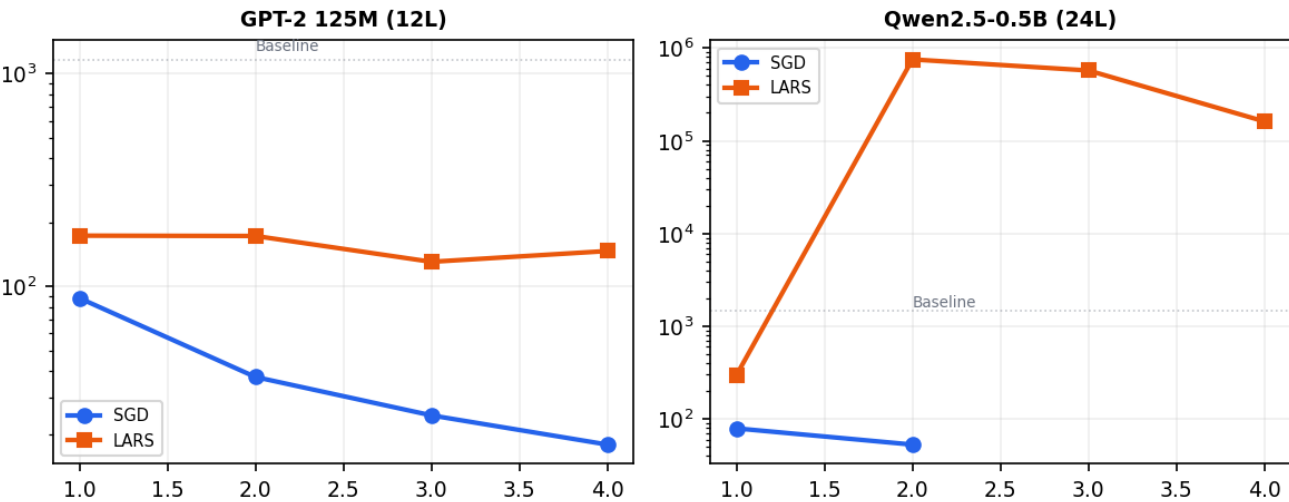


Fig 4: LARS vs SGD on GPT-2 12L (left) and Qwen0.5B 24L (right). LARS avoids NaN but does not converge.

4. A-SYNC Variant Progression

The path from the first A-SYNC to the final converging variant:

#	Variant	Key Change	7B Final PPL	Status
1	A-SYNC +perturb	Gradient inject + noise	25.8	Oscillates
2	A-SYNC no-perturb	Remove perturbation	16.6	Monotonic
3	A-SYNC 16-cosine	Cosine sync decay	10.5	Good, sync dies early
4	A-SYNC 32-cosine	Cosine over 32 cycles	13.3	Decay kills tail
5	A-CYCLE restart	3x8 warm restart	16.5	Window too short
6	A-SYNC 24-const	Constant sync, 24c	9.0	Excellent
7	A-SYNC 48-const	Constant sync, 48c	7.6	BEST — converged

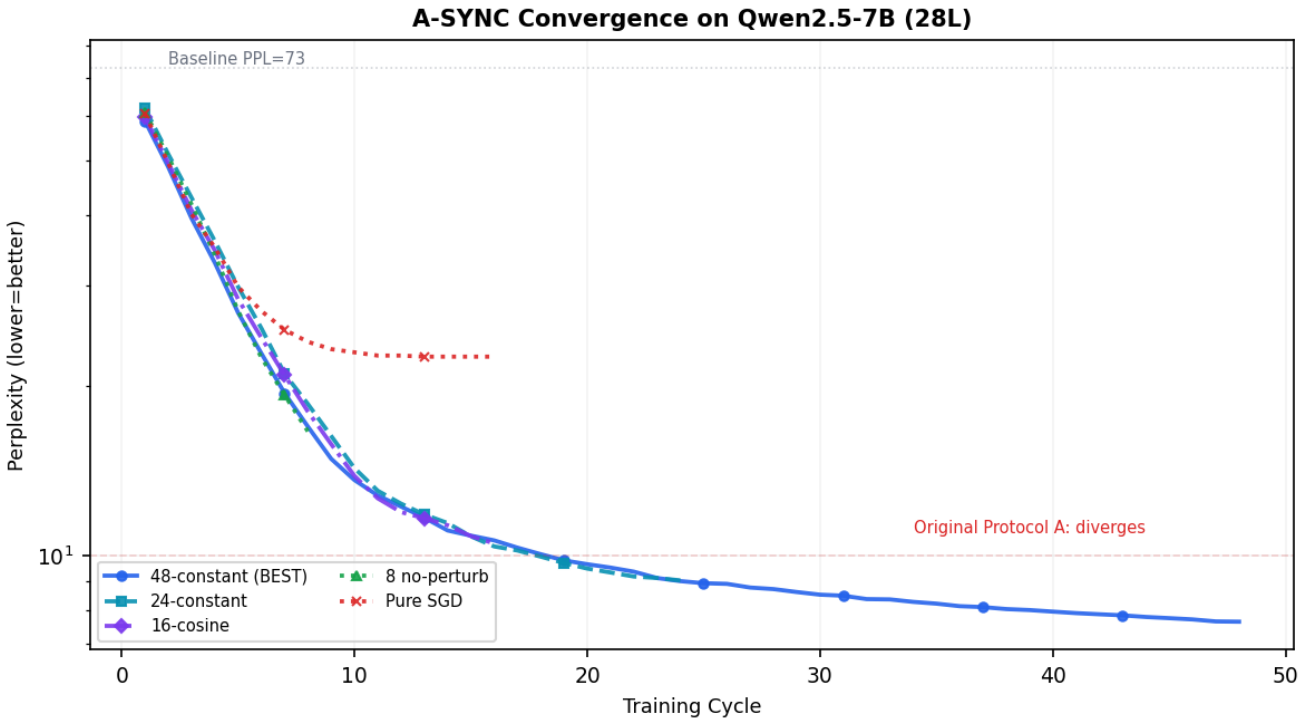


Fig 5: A-SYNC convergence curves on Qwen2.5-7B (28L). 48-cycle constant sync (blue) is the clear winner.

Key discovery: constant sync strength (0.05) dominates all decay schedules. The ALS signal should persist at full strength —

5. Final Scoreboard

Approach	
Category	
Qwen7B (28L) PPL	
Status	
A-SYNC 48-constant	
Gradient Injection	
7.6	
CONVERGED	
Pure SGD	
Baseline	
22.5	
Plateaus	
A-PROBE	
Architecture	
22.8	
Bottleneck limited	
A-KD	
Distillation	
195	
KL explosion	
LARS	
Optimizer	
161k	
No convergence	
Multi-layer ALS	
Algorithm	

- DIVERGED
- Underdetermined
- Parameter Tuning
- Tuning
- DIVERGED
- Non-transferable
- Protocol A (original)
- Baseline
- DIVERGED
- Residual amplification

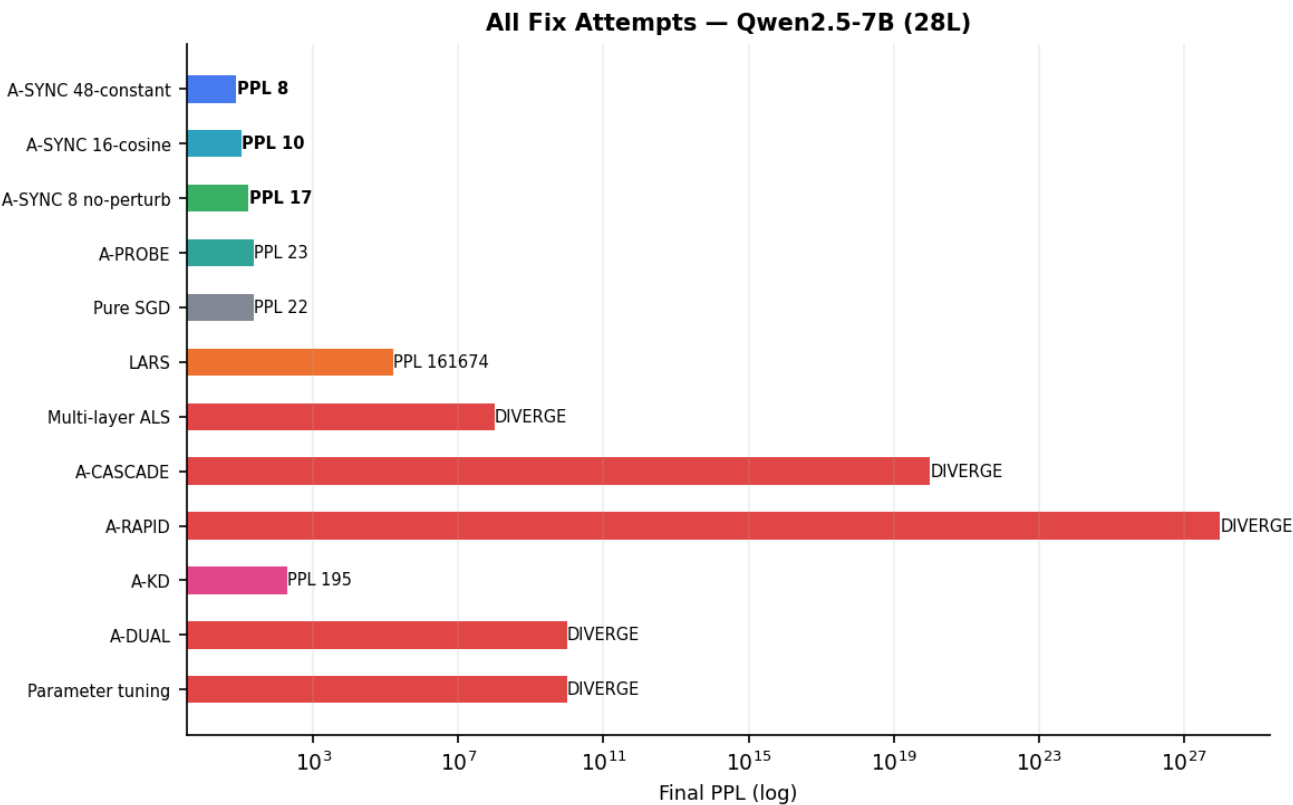


Fig 6: All fix attempts scoreboard — Qwen2.5-7B (28L). A-SYNC is 3x better than pure SGD.

Depth Boundary: Before vs After

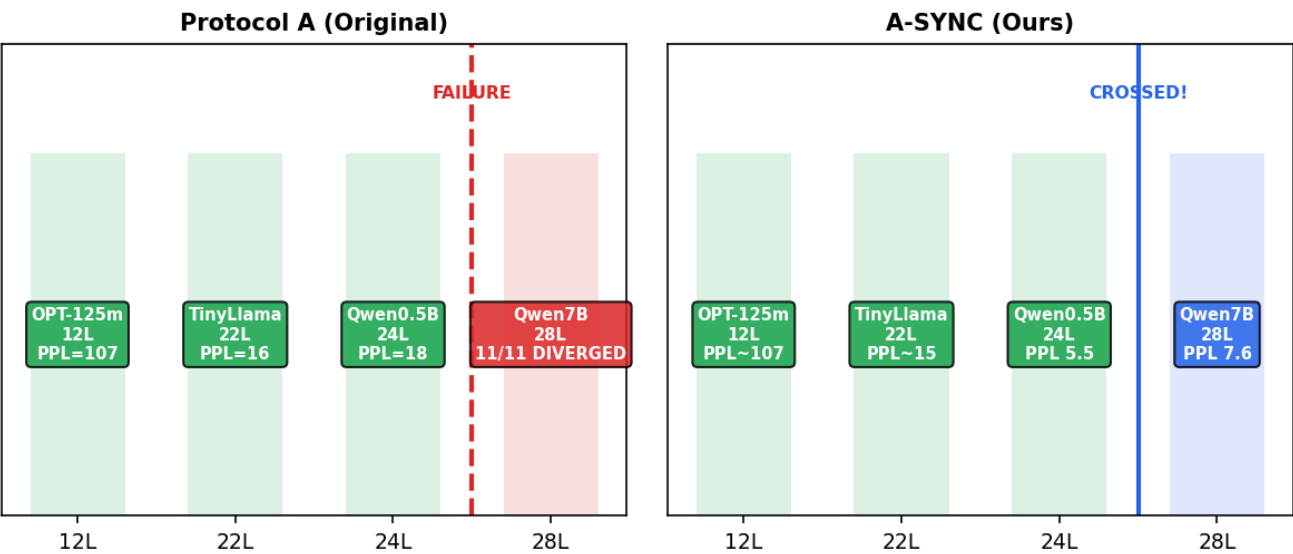


Fig 7: Protocol A (left) diverges at 28L. A-SYNC (right) crosses the boundary — monotonic convergence at all depths.